

SAKTHI VEL C

+91 6374052055 | c.sakthivel1.3.2006@gmail.com
linkedin.com/in/sakthivel-c | github.com/sakthivel-136 | sakthi-dev.in

PROFESSIONAL SUMMARY

Specializing in Machine Learning and IoT, with hands-on experience building and deploying production-grade AI applications using Gemini 1.5 Pro, FastAPI, and Next.js. Skilled in optimizing database performance and architecting secure, scalable full-stack systems, with a proven track record of independently shipping projects into real-world production use.

EDUCATION

Bachelor of Engineering in Computer Science

2023 – 2027

Kamaraj College of Engineering and Technology, Virudhunagar, India | GPA: 7.44 (till 5th sem)

TECHNICAL SKILLS

Python • SQL • FastAPI • Flask • Next.js 14 • PostgreSQL • Supabase • Tailwind CSS • RESTful APIs • JWT • Flutter • Git • GitHub • RBAC • TensorFlow • XGBoost • Random Forest • Gemini 1.5 Pro • ESP32/Arduino • MQTT

INTERNSHIP EXPERIENCE

Software Developer Intern

Dec 2025 – Jun 2026

PENTAGON GARMENTS, Virudhunagar, India

- Developed a secure verification system with RESTful APIs and JWT authentication for real-time security monitoring.
- Independently built and deployed the DigiSphere system into production for the company, currently in active daily use by their security staff.

Machine Learning Intern

June 2025 – Nov 2025

PANITH INNOVATIONS, Remote

- Developed and deployed predictive Machine Learning models using XGBoost and Random Forest to enhance system accuracy and project performance.

PROJECTS

DigiSphere | FastAPI, Next.js, PostgreSQL, JWT

July 2026

PROBLEM: Pentagon Garments had no digital way to verify security rounds — checks were logged manually on paper, making it hard to confirm rounds were completed or audit compliance.

SOLUTION: Built a full-stack verification system with FastAPI, Next.js, and RBAC for real-time monitoring, optimized queries to cut response time by 40%, and deployed it into live production for daily use by the security team.

Live In: apps.pentagontextiles.com

KURAL – AI Voice-Based Grievance Redressal System | Gemini 1.5 Pro, PSTN

Oct 2025 – Present

PROBLEM: Many people in rural and semi-urban areas face genuine grievances but lack the digital literacy or internet access to use complaint apps or web forms, leaving valid issues unreported.

SOLUTION: Built KURAL, a PSTN-based platform where anyone can report an issue with just a phone call — no app or internet needed. Gemini 1.5 Pro handles speech recognition and intent detection in real time, converting the call into a structured digital report and routing it to the right authority.

IoT Smart Air Quality & Climate Monitoring System | ESP32, Random Forest

Nov 2025

PROBLEM: Indoor air quality and HVAC inefficiencies often go unnoticed until they cause health issues or high energy costs, due to a lack of real-time monitoring.

SOLUTION: Built an ESP32-based sensor network feeding data into a Random Forest model to predict health risks and inefficiencies early, with findings presented at the ICASET-2026 International Conference.

Github Link: github.com/sakthivel-136/iot-enviroment-multi.git

AWARDS

- Grand Winner** – Recognized as the top team at the Mini Technovision 2K26 student pitch competition for an innovative solution pitch.

ACHIEVEMENTS

- Filed a patent application for "Mr. Strict", an AI-based proctoring system, to protect original IP for exam integrity technology.

CERTIFICATIONS

- Oracle Certified Foundations Associate** – Certified in Agentic AI and foundational technologies.
- ISRO Certification** – Completed AIML training for Crop Acreage Mapping via official ISRO workshop.
- Cisco Certification** – Completed a 70-hour course covering networking fundamentals and security protocols.
- Industry Certification** – Certified by Pentagon Garments for developing and deploying the DigiSphere system into real-world production use.